

NAAN MUDALVAN PROJECT

LegalEase: AI-Powered Legal Document Generator

Research and Select the Appropriate Generative AI

Project Overview

LegalEase is an AI-powered application designed to simplify the creation of commonly required legal documents. The project uses Generative AI to understand user requirements and produce structured document drafts while maintaining a clear, professional format. The system is intended to reduce repetitive drafting work, improve accessibility, and provide a guided document-generation experience for users.

Purpose of the Research

The research phase identifies the Generative AI technology that is most suitable for LegalEase. The selection must consider natural-language understanding, document generation quality, consistency, API availability, integration with a web application, response speed, scalability, security, and overall project feasibility. Since legal documents can have significant consequences, the application should present generated content as a draft for review rather than as a substitute for professional legal advice.

Research Objectives

1. Study Generative AI models capable of understanding detailed user instructions.
2. Compare model capabilities relevant to structured legal-document generation.
3. Identify a model that can be integrated efficiently with the LegalEase application.
4. Design a responsible workflow that includes validation, review, and user confirmation.

Expected Outcome

The research should result in a justified technology choice and a practical integration plan for the LegalEase prototype. The selected model should support reliable text generation, structured prompts, and application-level controls.

2. RESEARCHING GENERATIVE AI OPTIONS

Candidate technologies and selection criteria

Candidate Model Categories

The research can consider modern large language models available through application programming interfaces. Examples include Gemini-family models, GPT-family models, Claude-family models, and suitable open-source models. The final choice should be based on documented capabilities, current API support, project requirements, cost, latency, and responsible-use considerations rather than on model name alone.

Key Selection Criteria

- 1. Natural-language understanding:** The model should correctly interpret document type, user details, purpose, tone, and formatting requirements.
- 2. Structured generation:** It should produce predictable sections such as title, parties, clauses, dates, terms, and signature areas when required.
- 3. Instruction following:** The model should follow carefully designed prompts and avoid adding unsupported facts or unnecessary clauses.
- 4. API integration:** A clear SDK/API, authentication mechanism, documentation, and developer support are important for implementation.
- 5. Performance and scalability:** Response time, token limits, rate limits, and expected usage should be considered for the prototype and future expansion.
- 6. Safety and reliability:** The system should include safeguards against fabricated legal references, misleading statements, and inappropriate document generation.

Research Method

A practical evaluation can be performed using the same set of sample prompts across candidate models. Each output can be checked for completeness, clarity, structure, instruction adherence, and unsupported claims. The results can then be recorded in a comparison sheet and reviewed before selecting the model for the project.

3. COMPARATIVE EVALUATION AND SELECTION

Evaluation framework for LegalEase

Comparison Matrix

Criterion	What LegalEase Needs	Evaluation Focus
Language understanding	Accurate interpretation of user requirements	Prompt-following and context handling
Document quality	Clear, coherent, professional draft structure	Completeness and readability
Structured output	Consistent headings and fields	JSON/structured response support where applicable
API/SDK	Easy integration with backend	Documentation, authentication, SDK availability
Performance	Responsive user experience	Latency and usage limits
Scalability	Support for future users	Quota, rate limits, deployment model
Safety	Reduced risk of unsupported legal claims	Guardrails, validation and human review

Recommended Project Direction

For a Naan Mudalvan prototype, a managed Generative AI API is generally practical because it reduces the amount of infrastructure required to operate a large language model. Gemini-family models are a suitable candidate when the project is already designed around the Gemini ecosystem. The final selection should be confirmed against the current model documentation, pricing, quotas, supported SDKs, and institutional project requirements at the time of implementation.

Why This Choice Fits LegalEase

LegalEase needs an AI model that can transform structured user inputs into readable document drafts. A modern Generative AI API can provide the language-generation layer while the application controls forms, validation, templates, authentication, storage, and document export. This separation makes the project easier to test and maintain.

Selection Principle

The model should be selected through evidence-based testing rather than assuming that one model is universally best. The evaluation should use representative LegalEase prompts and measurable criteria.

4. IMPLEMENTATION PLAN AND CONCLUSION

Integrating the selected Generative AI into LegalEase

Proposed Workflow

1. **User Input:** The user selects a document type and enters the required information through a guided form.
2. **Validation:** The application checks required fields, formats, and missing information before sending a request.
3. **Prompt Construction:** Backend logic converts the validated information into a controlled prompt containing document requirements and formatting instructions.
4. **Generative AI Request:** The selected model generates a draft based on the supplied information.
5. **Post-processing:** The application checks the response for required sections, unwanted placeholders, and formatting issues.
6. **User Review:** The user reviews and edits the draft before export or use.
7. **Document Export:** The approved draft can be converted into a downloadable document such as PDF or DOCX.

Responsible AI Measures

LegalEase should clearly communicate that generated documents are drafts and that users should obtain qualified legal review when appropriate. The system should avoid inventing case numbers, laws, citations, identities, or factual details. Sensitive information should be handled carefully, and application logs should avoid unnecessary storage of private legal data.

Testing Plan

Testing should cover normal inputs, incomplete inputs, unusual wording, long inputs, invalid fields, repeated requests, and document-specific requirements. Generated documents should be evaluated for completeness, formatting, consistency, and unsupported information. API failures and rate-limit conditions should also have clear fallback messages.

Conclusion

The research and selection of an appropriate Generative AI model is a key foundation of the LegalEase project. A suitable model can provide the language-generation capability, while application logic provides validation, templates, security, review, and document export. By combining Generative AI with controlled workflows and human review, LegalEase can demonstrate a practical and responsible approach to AI-assisted legal-document drafting.

Project Deliverable: Research report and model-selection basis for the Naan Mudalvan LegalEase project.